Impacts of Public Early Childhood Education in Chile on Maternal Fertility and Labor Market Outcomes

Short-run effects — progress update

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Where we are

What we do

Question: does opening a public preschool nearby change mothers' labor supply and fertility?

Variation: 447 neighborhood units (UVs) open their *first* JUNJI/Integra preschool in 2015–2024 vs. 4,145 that never do

Design: staggered event study at the UV level, Sun–Abraham estimator (Baker et al., *JEL* 2026)

Data: woman-level panel linking administrative registries — RSH/FPS, SII/AFP earnings, Civil Registry births, preschool enrollment

What we find so far

First stage: enrollment **+2pp** (33% → 35%), from the opening year, no pre-trends; strongest at ages 2–3

Formal employment: **+1pp** from year 4 on; pre-trends not significant but rising

Earnings (mothers who work): **suggestive +2–4%**, not yet significant

Months worked: no effect so far

Fertility: no effect so far — a fairly precise zero

Cleaner control group: same results

Project status

Done: data, treatment, balance, first stage, first labor and fertility estimates

In progress: 2015 cohort check, earnings including non-working mothers

Next: pre-trend robustness, mechanisms, writing

Research question

Question

What are the effects of the public expansion of early childhood education in Chile on mothers' labor market participation and fertility?

What we know

Mixed evidence: positive effects on maternal employment in Quebec and Argentina, close to zero in Norway

Recent Latin American evidence comes from **lotteries and experiments** (Rio de Janeiro, Nicaragua)

Less is known about a **national, decade-long public expansion** observed in administrative data

Our contribution

First study to use neighborhood-unit (UV) variation: preschools geolocated to the UV and linked to individual administrative records

Fertility as a second margin, alongside labor supply

What is a UV? *Unidad Vecinal*: the official sub-municipal territorial unit — ∼6,900 UVs vs. 346 municipalities.

⇒ **Finer geography: a sharper measure of exposure and more variation to identify the effect.**

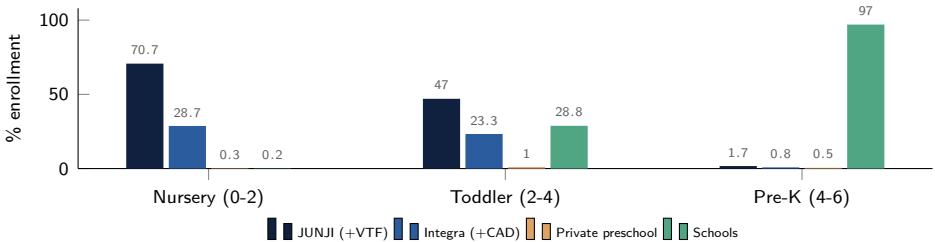
Quebec: Baker et al. (2008); Argentina: Berlinski & Galiani (2007); Norway: Havnes & Mogstad (2011); Rio: Attanasio et al. (2025); Nicaragua: Hojman & López Bóo (2022).

Public provision expanded sharply between 2015 and 2024

Mixed provision: **public** (JUNJI, Integra, municipal/SLEP) vs. **private** (subsidized/paid)

700+ new public preschools opened between 2015 and 2024

Net attendance rate (ages 0–5) rose from **50% to 55%** over the period, despite pandemic-era setbacks



Source: authors' elaboration based on Figura 5, Matrícula Oficial 2023, Centro de Estudios MINEDUC.

⇒ **Ages 0–4: JUNJI and Integra are the main providers — our focus.**

Identification: staggered-adoption event study

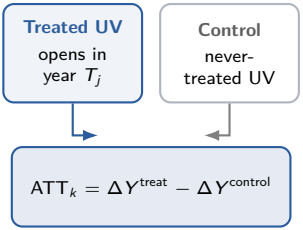
Design: staggered-adoption difference-in-differences with event study

Unit of treatment: the UV, the year it opens its first public preschool

Unit of analysis: **UV** $\times$ **year** panel, built from a woman $\times$ year panel linked to each woman's UV of residence

Fixed effects: **UV + year** (baseline); woman FE on the woman $\times$ year panel as a **robustness check**

Estimand: ATT of UV-level treatment (Baker et al., *JEL* 2026) — an **intent-to-treat effect of access**, not individual take-up



$\Rightarrow$ **We compare changes in UVs that get their first preschool vs. UVs that never do — an intent-to-treat effect of access.**

A UV is treated when it opens its first preschool

Treatment definition

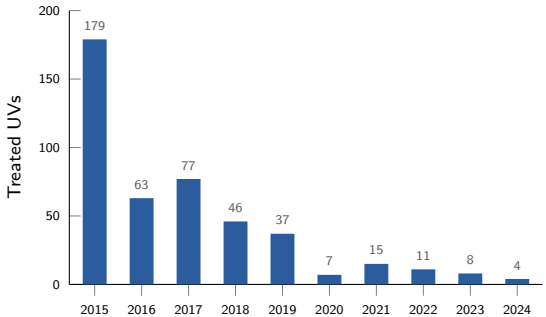
A UV is **treated** the year it opens its first public preschool (JUNJI/Integra) — goes from 0 to 1+ centers.

Group	Definition	N (UVs)	%
Excluded	Already had 1+ center in 2014 (out of design)	2,265	33.0%
Never-treated	0 in 2014, stays at 0 through 2024	4,145	60.5%
Treated	0 in 2014, opens first center 2015–2024	447	6.5%
Total universe		**6,857**	100%

Source: `base_cobertura_cp.dta`. Design sample (`muestra_t1==1`) = Never-treated + Treated = 4,592 UVs.

$\Rightarrow$ **We identify the effect of the first preschool ($0 \rightarrow 1$): 447 treated vs. 4,145 never-treated.**

Treatment cohorts by year



Source: `base_cobertura_cp.dta`. 447 treated UVs (0 centers in 2014).

2015 cohort is the largest: 179 UVs (40%), tied to the Bachelet preschool expansion

71% open in 2015–2017; only 45 UVs open after 2019

2014 as base year lets us identify the 2015 cohort as $0 \to 1$; without it, treated UVs drop from 447 to $\sim$268

But the 2015 cohort has **only one pre-period** and sits at the edge of the panel $\to$ we come back to it below

$\Rightarrow$ **Most openings are early (2015–17): long post-treatment windows for most treated UVs.**

Empirical specification and estimator

$$Y_{ut} = \lambda_u + \lambda_t + \gamma\, \overline{\text{Age}}_{ut} + \sum_{k \neq -1} \beta_k\, \mathbf{1}\{t - G_u = k\} + \varepsilon_{ut}$$

Y_{ut}: average outcome of women in UV u, year t (built from the woman $\times$ year panel). λ_u, λ_t: UV and year fixed effects; SE clustered by UV.

$\overline{\text{Age}}_{ut}$: average age of mothers in the UV-year (labor market specification).

β_k: effect k years from opening, relative to $k = -1$ (omitted); $k \leq -5$ binned.

Estimation strategy

Baseline TWFE as reference — but with staggered treatment, TWFE can use already-treated units as controls (Goodman-Bacon problem). Robust estimator: **Sun–Abraham** (`eventstudyinteract`), never-treated UVs as control group. Framework: Baker, Callaway, Cunningham, Goodman-Bacon & Sant'Anna (*JEL*, 2026).

Main threat: non-random placement of openings (JUNJI/Integra prioritizing UVs by care demand or socioeconomic profile). **UV FE** absorb permanent differences across UVs; **pre-period coefficients** ($k < -1$) test whether trends were already diverging.

$\Rightarrow$ **Main estimator: Sun–Abraham with never-treated UVs as controls; TWFE as reference.**

Data: what we have and how each sample is built

Sample	Source data	Key variables
Coverage (UV × year)	JUNJI/Integra transparency requests + UV shapefile (MDSF)	Number of centers, treatment status, opening cohort
Women panel (base)	RSH (2016–2024) + FPS (2014–2015)	UV of residence, UV population, child density
Labor market	+ Rentas (SII/AFC) + Cotizaciones (AFP)	Annual income, months worked, employment
Fertility	+ Civil Registry birth records (via RIS)	Birth in the year, children under 5
First stage (enrollment)	+ Matrícula parvularia (child-level)	Enrollment, enrollment rate

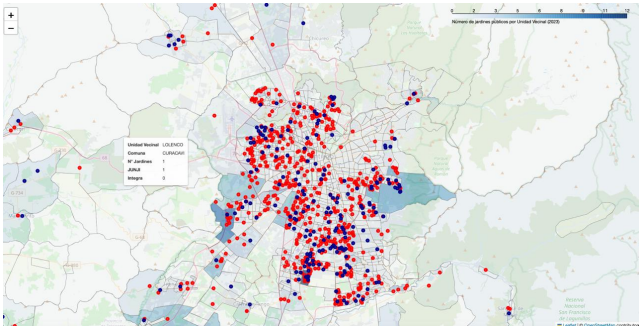
⇒ **All administrative registries, linked at the individual level — no survey self-reports.**

Every preschool is matched to its 2024 neighborhood unit

Each JUNJI/Integra preschool is **geolocated** and matched to the UV where it sits (spatial join)

UV boundaries **changed a lot** over time → we traced them and mapped everything to **2024 boundaries**

Preschools and women's UV of residence share **one common geography** across all years

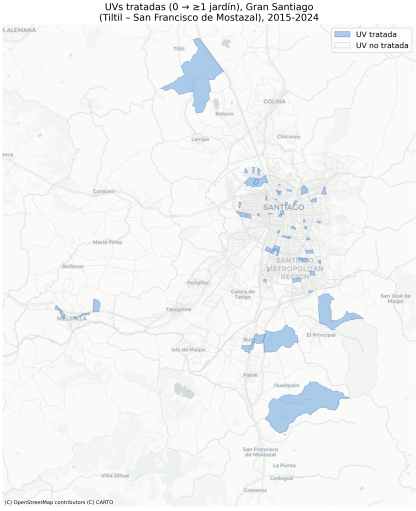


Región Metropolitana. ● JUNJI ● Integra Shading: public preschools per UV. Source: `base_cobertura_cp.dta`.

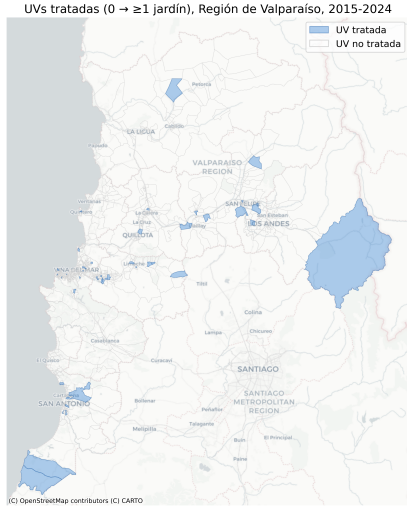
⇒ **One common geography: every preschool and every woman mapped to 2024 UV boundaries.**

▸ Interactive map 2014–2024

Región Metropolitana



Valparaíso



Illustrative examples. Shaded: treated UVs (0 → 1+ preschools), 2015–2024.

Who is in each sample, and what we measure

1. Samples (selected at the individual level)

Fertility: women of fertile age (18–49)

Labor market: women with a child <5 *that specific year* — not "ever had a child"

First stage: children of treatment-sample mothers, birth to birth+4

2. Outcomes (averaged at UV $\times$ year)

Fertility: share of women who had a child that year (Civil Registry birth records)

Labor market: average annual income, months worked, share employed, dependent vs. independent

First stage: enrollment rate

$\Rightarrow$ **Each outcome is measured on its relevant population, from administrative records, averaged at the UV $\times$ year level.**

Treated and never-treated UVs look alike — except in size

Labor market sample (mothers with a child <5) — covariate levels in 2014, before any opening

	Unweighted			Weighted by UV population			
Variable	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	Sig.[a]
Annual income (M CLP)	3.60	3.55	-0.033	3.95	3.89	-0.039	*
Months worked	8.14	8.10	-0.024	8.44	8.37	-0.058	*
Age	29.24	29.25	0.008	29.26	29.29	0.026	
N. children <5	1.11	1.11	0.045	1.11	1.11	0.049	
Hours worked	37.07	37.68	0.099	36.55	36.81	0.068	
Works (share)	0.45	0.45	0.023	0.48	0.48	-0.006	
Education (code)	4.98	5.02	0.028	5.20	5.18	-0.020	
UV population	1126.79	2011.08	**0.410**	3530.92	5287.26	**0.340**	***
Child density 0–5 UV	0.06	0.06	0.155	0.06	0.06	0.139	***
N (UVs)	3,909 / 422			3,909 / 422			

[a] t-test at the woman level (* $p<0.05$, *** $p<0.01$), for reference only; normalized differences prioritized (concern threshold: 0.25).

The UV population gap is **expected**: the State prioritizes new preschools where need and population are larger. Same pattern in the fertility sample (App. A4).

⇒ **Selection is in *where* preschools open (UV size), not in *who* lives there → UV fixed effects.**

About 1 in 3 mothers moves across UVs — equally in treated and control UVs

How many move?

33% of mothers change UV at least once while in the sample

7–12% per year

2022: 27% — likely a cartography artifact, not real moves

Do they move toward new preschools?

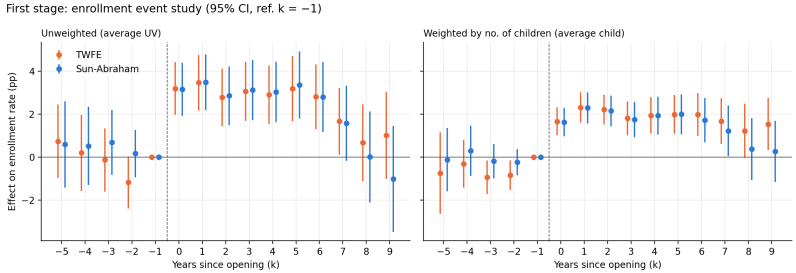
Group	Moves (% of mother-years)
Treated	12.6
Never-treated	12.3
Excluded	11.9

Moving is frequent: part of the mothers in a treated UV arrive after the opening and others leave → dilutes UV-level effects. We address it by following each mother from her first UV (woman fixed effects, below).

⇒ **Mothers move often, but not toward UVs that open a preschool.**

Moves measured with 2024 UV boundaries, the same map that defines treatment. Year-to-year changes from 2017 on (2014–2015 UVs are proxied by 2016). 33% = among mothers observed in at least two comparable years.

First stage: enrollment rises the year a preschool opens and stays higher



Weighted (main): ≈**+2pp** on a 33% base (≈+6%), immediate and stable for $k = 0$–6

Unweighted: ≈+3pp — larger UVs move less with the same new places

No pre-trends: small and not significant

$k = 7$–9: only early cohorts (mostly 2015) observed → composition, not a fading effect

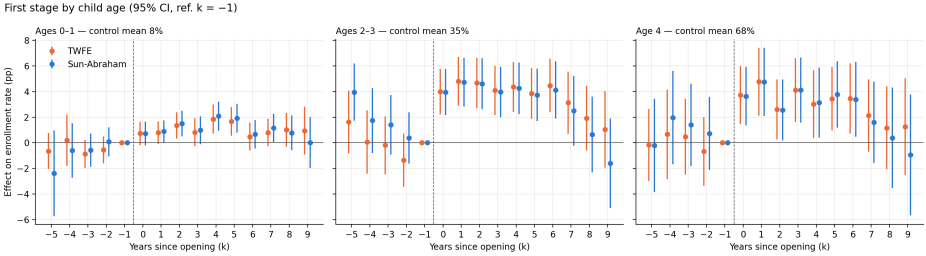
▸ Average effects (A6)

⇒ **A new preschool raises enrollment from 33% to 35%: about 3 more children enrolled per UV.**

Sun–Abraham and TWFE, never-treated UVs as controls, UV and year FE, 95% CI clustered by UV, reference $k = -1$.

Children aged 0–4.

First stage by child age: strongest at ages 2–3



Child age	Control mean	SA $k = 0$–6	Relative
Ages 0–1	8%	+1.3pp	+15%
Ages 2–3	35%	**+4.2pp**	+12%
Age 4	68%	+3.6pp	+5%

Ages 2–3: largest effect, immediate and stable

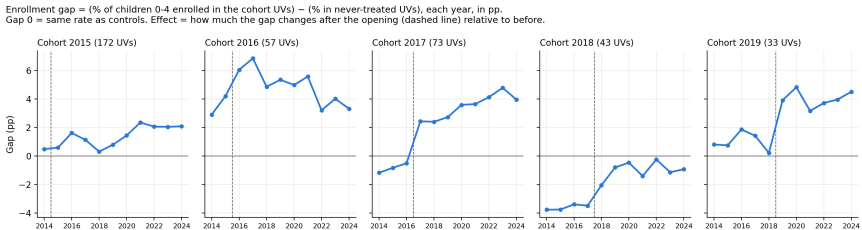
Ages 0–1: small in pp but largest in relative terms; grows over time

Age 4: also positive

Caution: ages are measured at different points in time across datasets (to check)

⇒ **Enrollment rises at all ages, most at ages 2–3.**

An open issue: the 2015 cohort



172 UVs — 41% of treated UVs, ∼half of the sample at $k = 6$, the only cohort at $k = 9$

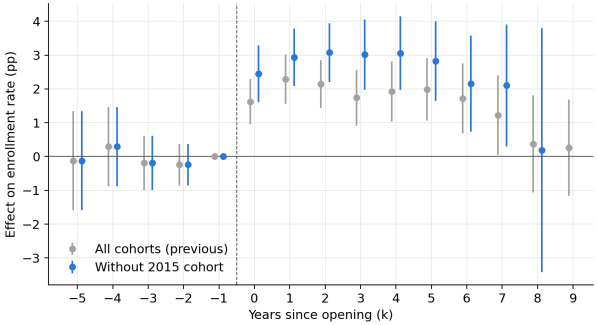
Its effect is **very small** (≈0.7pp in raw gaps) and it has only one pre-period (2014)

Hypothesis: registration error at the edge of the panel — preschools that already operated in 2014 but are recorded as 2015 openings.

Next: carefully review the coverage data provided by JUNJI/Integra.

⇒ **The 2015 cohort pulls the average effect down — possibly a data error to check.**

Without the 2015 cohort, the first stage is ∼50% larger



Same pre-periods (the 2015 cohort has none)

+2.7pp vs. +1.8pp: **≈4 more children** enrolled per UV instead of 3

Jumps at opening and stays stable

Earnings: the effect also roughly doubles without 2015

⇒ The 2015 cohort dilutes the first stage — but it remains small.

All mothers (first stage not yet estimated for the 60% most vulnerable). Average $k = 0$–7. Sun–Abraham, weighted by number of children, 95% CI clustered by UV.

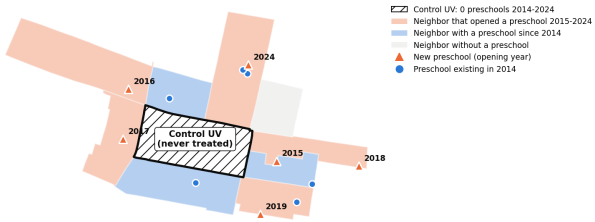
A cleaner control group: never-treated UVs can be exposed too

Concern: children from a never-treated UV can enroll in a new preschool next door → control enrollment rises → **effect underestimated**

We built a **neighbor map**: for every UV, which adjacent UVs had or opened a preschool (shared border, 2024 boundaries)

Example: this control UV has 9 neighbors — **5 open a preschool** during the period

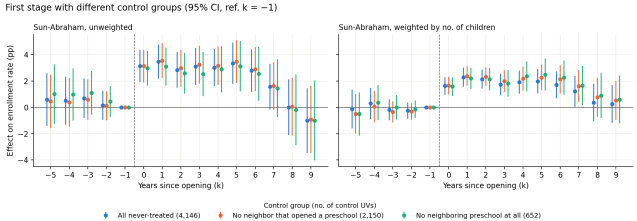
A 'never-treated' UV in La Cisterna
5 of its 9 neighboring UVs open a preschool in the period



⇒ **Goal: a larger first stage, comparing treated UVs with controls that could *not* use a neighbor's new preschool.**

* Neighbors only capture proximity to home. If mothers use preschools near their workplace, spillovers may reach UVs that are not adjacent.

A cleaner control group gives the same first stage



Control group	UVs	Median area	Control mean	SA weighted, $k = 0$–6
All never-treated (main)	4,146	2.8 km^2	33.2%	1.92
No neighbor opened a preschool 2015–24	**2,150**	3.8 km^2	32.7%	**2.14**
No neighboring preschool in any year	652	**6.7 km^2**	28.2%	2.13
Treated UVs	*422*	*1.5 km^2*		

Same effect with all three groups and clean pre-trends. The strictest group is not a good comparison: its UVs are $\sim$**4**$\times$ **larger** than treated ones (isolated areas) $\rightarrow$ preferred clean control: **no neighbor opened**.

$\Rightarrow$ **Spillovers to controls do not explain the small first stage.**

Why is the first stage small?

What we checked

2015 cohort: very small effect, possibly a data error → without it, ≈3pp

Cleaner control group: same ≈2pp → spillovers to controls are not the reason

What our outcome measures

Whether children **living** in the treated UV are enrolled in **any** preschool

A new preschool can fill up without moving our outcome

It raises enrollment only if it takes local children who were **not enrolled before**

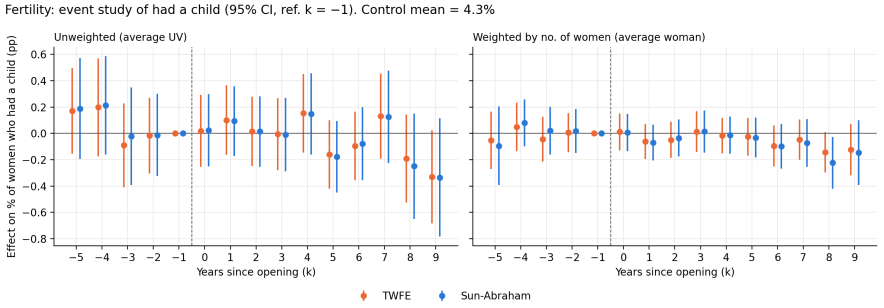
Substitution: local children already enrolled elsewhere switch to the new preschool → no net change

Mothers use other preschools, e.g. near their workplace in another UV → new places go to children from other UVs

Future work: enrollment records show which preschool each child attends → who fills the new preschool, where they live, and which preschools local mothers use.

⇒ **Enrollment of local children can move little, even if the new preschool fills up.**

Fertility: no effect yet



Treated and control UVs decline together; **no jump at opening**

Clean pre-trends

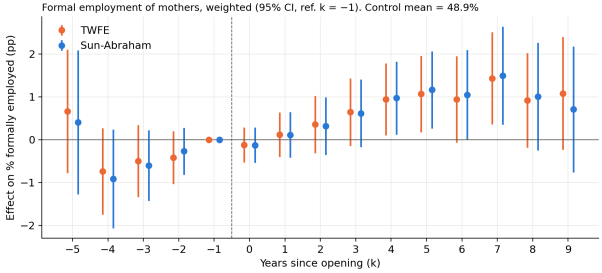
A **fairly precise zero**: rules out large effects

Still checking: cumulative outcome (having had a child)

⇒ **Expected: having a child depends on many factors, and a nearby preschool might weigh little.**

Women aged 18–49. Outcome: share who had a child in the year (control mean 4.3%). Controls for average age.

Mothers' formal employment rises by 1pp from the fourth year



Nothing at opening; **significant every year from $k = 4$ to 7**

Why gradual?

Each year's sample = mothers with a child <5

At $k = 0$, many children are already 3–4: little time to use the preschool

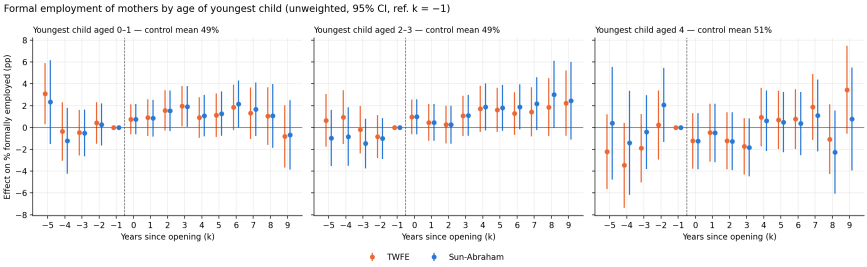
From $k \approx 4$, every mother in the sample had it available **since her child was born** → exposure builds up

Caution: pre-trends not significant (joint test $p = 0.82$) but slightly rising; pre and post years come from different cohorts.

⇒ A new preschool raises mothers' formal employment by ≈1pp from the fourth year.

Weighted by number of mothers. Mothers with a child under 5. Employed = any taxable income or social security contribution in the year.

Employment effect appears for mothers whose youngest child is 0–3



Youngest child **0–1 and 2–3**: positive, +1 to +2pp

Youngest child **4**: no effect

Fits the mechanism: mothers of younger children face a tighter care constraint; at age 4, most children are already enrolled (68%)

Not significant one by one — the **pattern across groups** is what is informative

⇒ **The effect comes from mothers of children aged 0–3, who need care the most.**

Unweighted (weighted version not yet estimated).

Changes to the samples

1. Only the 60% most vulnerable (labor and fertility samples)

The population **eligible** for JUNJI/Integra preschools → removes women who could not be treated

FPS score ($\leq$13,500) in 2014–2015, RSH bracket ($\leq$60) from 2016; **fixed per woman** at her first year with data

2. Mothers stay in the labor sample (in progress)

Before: a mother is in the sample only **while** she has a child <5

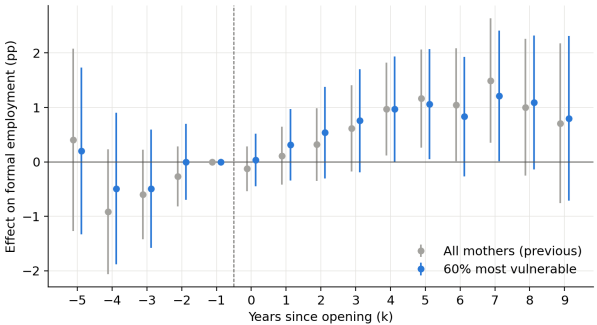
Now: she enters the first year she has a child <5 and **stays until 2024**, even after the child turns 5

The sample no longer changes year to year with births and children aging out

⇒ **We now focus on eligible mothers and follow the same mothers over time.**

Results below use change 1 with the previous mother definition; change 2 is being built.

Employment: same effect among the 60% most vulnerable



No pre-trends (joint test $p = 0.79$)

Gradual: nothing at opening, ≈**+1pp from the fourth year**

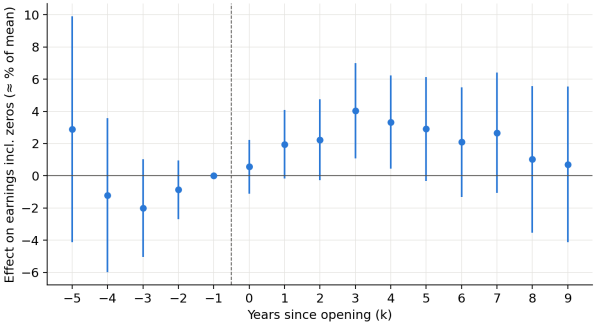
Average after opening: **+0.8pp** on a 46% base (≈+1.6%), significant

Almost identical to all mothers: vulnerable mothers are 2/3 of the sample

⇒ **The employment effect holds in the eligible population.**

Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV, reference $k = -1$. Mothers with a child <5 that year.

Earnings rise because more mothers work, not because they earn more



Earnings **including zeros**: average over *all* mothers, those without earnings count as 0

Y-axis: 100 = average mother that year → **+2 points ≈ +2%**

Average after opening: ≈**+2%**, peak at $k = 3$–4 (significant at 10%)

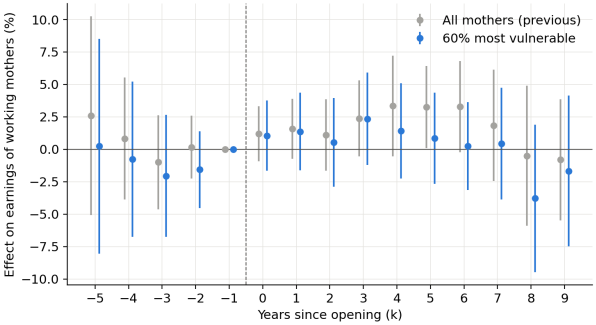
∼**3/4** of it is explained by more mothers working

Among mothers who work: earnings and months worked **do not change**

⇒ **The preschool changes *whether* mothers work, not how much those already working earn.**

60% most vulnerable. Earnings relative to the year's mean (100). Sun–Abraham, weighted by number of mothers, 95% CI clustered by UV.

Working mothers do not earn more



In earnings: **only mothers who work**

No pre-trends; after opening, ≈**0** among vulnerable mothers (+0.3%, not significant)

All mothers: slightly higher (≈+3% at $k = 3$–6), also not significant

Months worked: also no effect

⇒ **The earnings gain does not come from higher wages.**

Sun–Abraham, weighted by number of mothers with earnings, 95% CI clustered by UV, reference $k = -1$. Mothers with a child <5 that year.

Following each mother: woman fixed effects (preliminary)

Fixed UV

Each mother is assigned to the **first UV where she appears**; if she moves, she keeps that UV's treatment

Compares **the same mother** before and after the opening → removes fixed differences between mothers and the effect of moves

Result: formal employment

No pre-trends; gradual increase, **significant from the fourth year**

Same pattern as the UV-level estimate

⇒ **Following the same mothers confirms the employment effect.**

Disclaimer: for women first observed in 2014–2015, the UV in those years is proxied by their 2016 UV (App. A2), so their fixed UV is the 2016 one. Sun–Abraham, woman and year FE, clustered by fixed UV. To be re-estimated with the new mother sample.

Running and pending

Running

New samples: vulnerable mothers followed until 2024; fertility and first stage for the 60% most vulnerable

Woman fixed effects with the new sample: average effect and SE

Pending

All outcomes with the new samples

First stage among the 60% most vulnerable

Balanced event study (same cohorts pre and post)

The other 40% separately: is the effect concentrated among vulnerable mothers?

2015 cohort: were 2015 "openings" already operating in 2014?

Treatment by initial stock of preschools

Sample by predicted probability of pregnancy

Appendix

Appendix A1. Full identification assumptions and placebo check

Parallel trends: assessed via pre-opening event-study coefficients; should be statistically indistinguishable from zero

No anticipation: mothers do not adjust decisions before the effective opening — depends on how much advance notice openings receive (to analyze)

Exogeneity of timing: opening timing should not correlate with unobserved UV characteristics. **Main threat**: JUNJI/Integra may prioritize UVs with higher care demand or specific socioeconomic profiles

Placebo check

An analogous event study will be estimated using the **pre-treatment birth rate** of the UV as the outcome. Non-significant pre-treatment coefficients would strengthen the parallel-trends assumption of the main model.

Appendix A2. Data limitation: UV assignment in 2014–2015

FPS (2014–2015) has no reliable UV field → those two years are anchored to the woman's **2016 UV** as a proxy

Consequence: women who left the panel before 2016 are dropped from 2014–2015 — precisely the earliest pre-trend years

RIS has not delivered a better historical crosswalk despite repeated requests; only unstandardized UV name strings are available

Idea, not yet tested: fuzzy matching on UV name strings

Appendix A3. Task status

Block	Task	Status	Note
2. Data generation	Coverage base (preschools × UV, 2014–2024)	ready	Verified; private preschools (transparency) still a possible extension
	Women panel (2014–2024, incl. FPS anchor)	ready	2014–2015 anchored to 2016 UV (see A2)
	Rentas / Cotizaciones cleaning	advanced	Independent/state/FF.AA. cotizaciones still pending
	Children registry cleaning	ready	`tuvo_hijo`, `n_hijos_menor5`
	Merge panel + outcome variables	ready	By `rut_inn`
	Treated / non-treated definition	ready	`grupo_t1`, `g1`, `muestra_t1`
3. Panel analysis	First-stage estimation (enrollment)	ready	+3pp; 2015 cohort under review
	Balance / validity check	ready	UV-level balance table
4. Data analysis	Methodology choice	ready	Sun–Abraham within Baker et al. (JEL 2026) framework
	Treatment definition	ready	Extensive margin, 0→1
	Run regressions (labor, fertility)	ready	Re-run without 2015 cohort pending
	Analyze results	in progress	Pre-trend robustness for employment
5. Mechanisms	Heterogeneity by group / level	in progress	By child age (first stage, employment)
	Public-policy implications	not started	
6. Writing	—	in progress	

Appendix A4. Balance in the fertility sample

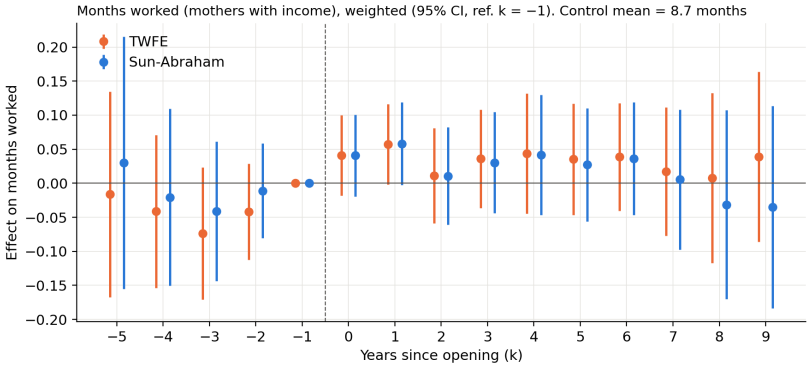
Fertility sample (women aged 18–49) — covariate levels in 2014, before any opening

	Unweighted			Weighted by UV population			
Variable	Never-tr.	Treated	Norm. diff.	Never-tr.	Treated	Norm. diff.	Sig.[a]
Birth rate	0.057	0.058	0.031	0.06	0.060	0.051	
Age	31.90	31.83	-0.053	31.74	31.82	0.095	
Children <5 per woman	0.303	0.319	0.156	0.31	0.320	0.117	**
UV population	1111.42	2025.96	**0.423**	3540.63	5305.76	**0.342**	***
Child density 0–5 UV	0.0590	0.0631	0.176	0.06	0.064	0.148	**
Women in sample per UV	233.60	445.45	**0.401**	795.29	1234.46	**0.347**	***
N (UVs)	3,973 / 422			3,973 / 422			

[a] t-test at the UV level (** $p<0.01$, *** $p<0.001$), for reference only; normalized differences prioritized (concern threshold: 0.25). UVs with at least one sample woman in 2014, hence more never-treated UVs than in the labor sample.

⇒ **Same conclusion as the labor sample: women look alike, UVs differ in size.**

Appendix A5. Months worked: no effect so far



Weighted. Mothers with earnings: about +0.03 months on a base of 8.7, clean pre-trends.

Caveat: currently conditional on working. Next: months worked with 0 for mothers without work.

Appendix A6. First stage: average effect is ≈2pp for the average child

Back to first stage

Sun–Abraham average (pp)	**Weighted (main)**	Unweighted
Post, $k = 0$–6	**1.92**	3.11 (0.58)
Post, $k = 0$–9	1.53	2.23 (0.63)
Pre, $k = -5$–-2	−0.06	0.49 (0.59)
Joint pre-test	—	$p = 0.90$
Control mean	33.2%	32.4%
Relative effect ($k = 0$–6)	**+6%**	+10%

Weighted = effect for the average child (weights: children aged 0–4 in the UV). **Unweighted** = effect for the average UV.

What changed since last meeting: the model now includes *all* event-time dummies. The negative $k = -2$ we flagged came from TWFE; it disappears with Sun–Abraham.

Average of event-study coefficients; SE (in parentheses) uses their full covariance matrix. Weighted SE pending. $k = 0$–6 is our main window (balanced across cohorts).

⇒ A clear first stage: 33% → 35%, immediate, no pre-trends — but small.

Appendix A7. Self-reported outcomes (RSH): exploratory

Outcome (mothers, RSH)	Control mean	Weighted	Unweighted
Not looking for work: no childcare	11.8%	−0.4pp	**−0.7pp***
Looking for work	18.6%	+0.6pp	+0.8pp
Weekly hours worked	36.5 h	+0.1 h	−0.3 h
Not looking for work: household chores	12.0%	not interpretable (pre-trends)	

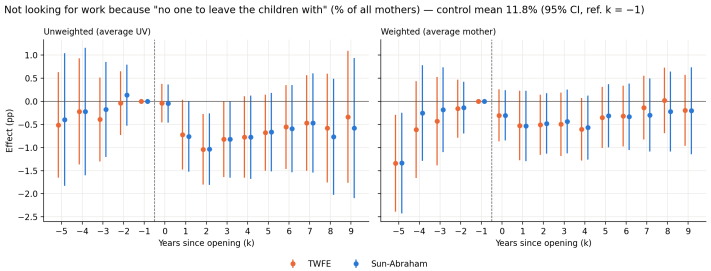
Most interesting: fewer mothers say they don't look for work because they have no one to care for their children (−6% relative), from $k = 1$, clean pre-trends — the most direct measure of the care-constraint mechanism

Weighted: same direction, weaker

Caveats: self-reported; the RSH form is not updated every year, and families may update it when applying for benefits (e.g., a preschool) → timing may depend on treatment

Sun–Abraham average effect, $k = 0$–6. * $p < 0.05$. Weighted SE pending.

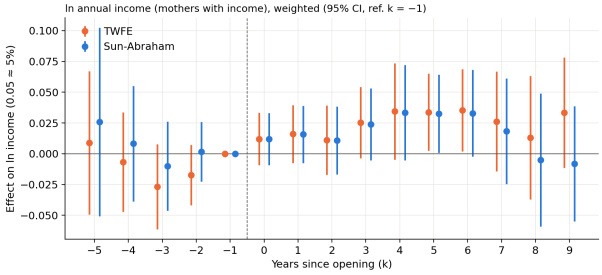
Appendix A8. RSH: fewer mothers lack childcare



Fewer mothers say they don't look for work because they have no one to leave the children with. Unweighted: -0.7pp (-6%), significant from $k = 1$, clean pre-trends; weighted: same direction, weaker

Caveats (exploratory): only mothers who neither work nor look for work answer the reason; all others are coded 0, so it is a % of all mothers — if more mothers start working, it falls too (part of the effect). The RSH form is not updated every year: the same answer can repeat for several years, so changes are slow and noisy. Some UVs have very few answers ($\sim$4 women).

Appendix A9. Previous earnings estimate (all mothers)



Same **gradual shape** as employment; clean pre-trends

$\approx$+3% at $k = 4$–6; average $k = 0$–6: **+2.3%** $\approx$ CLP 10–20k per month, not yet significant

Still to do

ln earnings is measured **only among mothers with earnings** $\rightarrow$ new entrants may pull it down

Add earnings **in levels, including zeros** for mothers without earnings

Months worked (App. A5): no effect so far, but also conditional on working $\rightarrow$ re-estimate with 0 for mothers without work

$\Rightarrow$ **Mothers who work earn $\approx$2–3% more from the fourth year: a suggestive result.**

Appendix A10. Two ways to measure earnings

Example: a UV with 4 mothers. After the opening, **C starts working**, at a somewhat lower wage (CLP million per year).

	A	B	C	D	**Earnings incl. zeros** (all mothers)	**ln earnings** (only mothers who work)
Before	4	4	0	0	$(4+4+0+0)/4 = \mathbf{2.0}$	mean of A, B $= 4$
After	4	4	**3**	0	$(4+4+3+0)/4 = \mathbf{2.75} \uparrow$	mean of A, B, C $\approx 3.6 \downarrow$

Earnings including zeros: how much income mothers have *overall* → rises if more mothers work *or* if they earn more

Expressed relative to the year's mean (100): +2 points ≈ +2% for the average mother

ln earnings: how much *working* mothers earn → can fall when new workers enter with lower wages, even if no one's wage falls

⇒ **Earnings including zeros capture new workers; ln earnings only captures wages.**